

Tools Used:

- Pandas
- Flask
- D3
- HTML, CSS, JS

Question 1: Characterize the data. What aspects of this data could serve as units of analysis? How do those components relate to one another?

Please limit your answer to 8 images and 500 words.

The dataset is a collection of stoppage records at train stations along the greenline. Every single entry corresponds to one individual stop. From this we get five different categories we can separate our stops:

- Stop
 - The station at which the train is stopping at. Certain stations such as Medford/Tufts belong to a specific branch, and some such as Copley are shared.
- Branch
 - The branch of the T that the train is running on, the Green Line Extension, corresponds partially to the D and E, whereas the B and C stop at Copley. For stops that serve multiple branches, the specific branch that the stopped train is serving is listed
- Direction
 - Whether the train is travelling eastbound or westbound, westbound being away from the Green Line Extension and eastbound being towards Medford/Tufts
- Time of day
 - The precise time of day, measured in seconds since midnight, when the train stops at the station. In our visual analytics system, this can be separated into buckets of customizable size.
- Date
 - The date of the trip, analysis of the date is slightly hindered by the fact that our dataset is not complete

The main aspect of this data that can serve as a unit of analysis is the headways data. The data is a collection of what are effectively timestamps of a train stopping at a station, giving info on the time, station, branch, route, direction, and the headway. The headway is the difference in seconds between a train and the next. It is measured both by the next train of any type, and the next train of the same branch, specified as headway_trunk_seconds and headway_branch_seconds. We have data for every month of 2025, although it is an incomplete dataset. The components of this

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data relate to each other in a hierarchy. Each stop belongs to a certain trip in a certain direction on a certain branch which is in turn a part of the green line. Headway could be implicitly calculated by viewing the previous stop from a separate train, or a train of the same branch exclusively if calculating branch headway, its timestamp, and the timestamp of the current train's stop. One of the more interesting facets of the data is that in certain cases, although not on the green line extension, the branch and trunk headway are not meaningfully different to a passenger as many stops are shared by branches, only splitting off later. Thus, we must interpret the difference between branch and trunk headway differently depending on the specific stop.

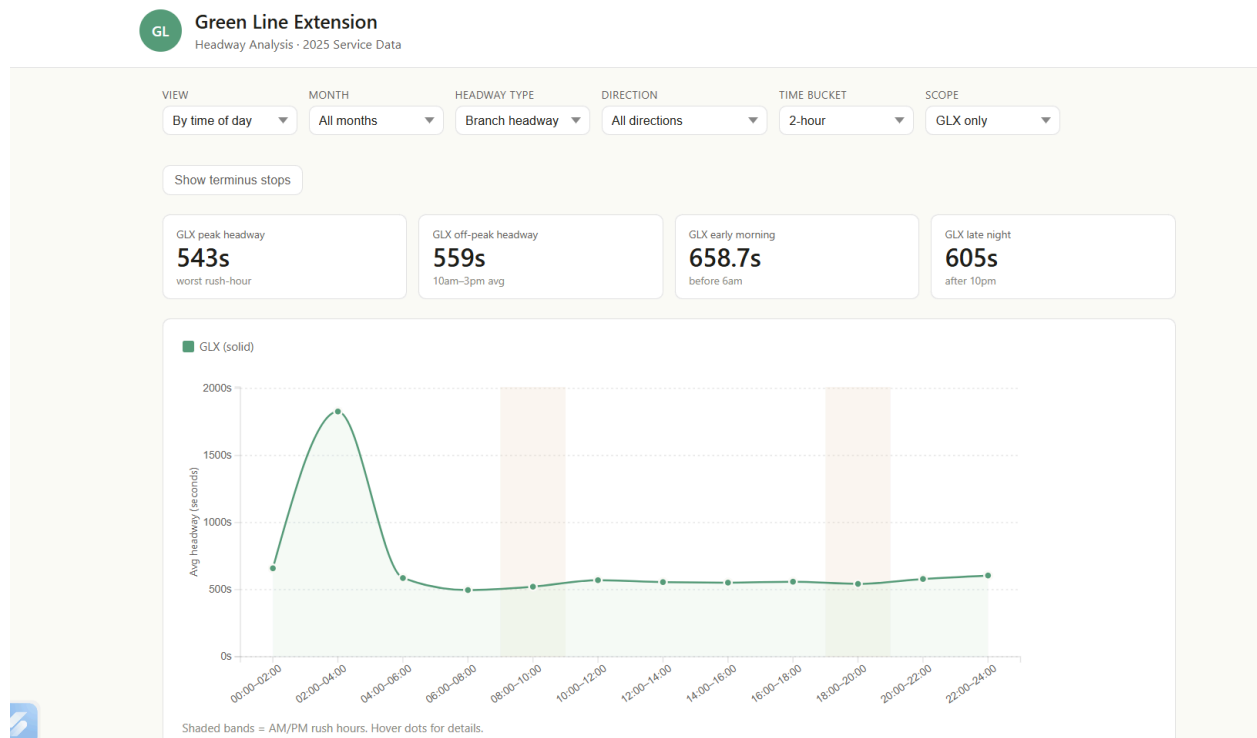


Figure 1: Average branch headway by time of day in 2-hour buckets

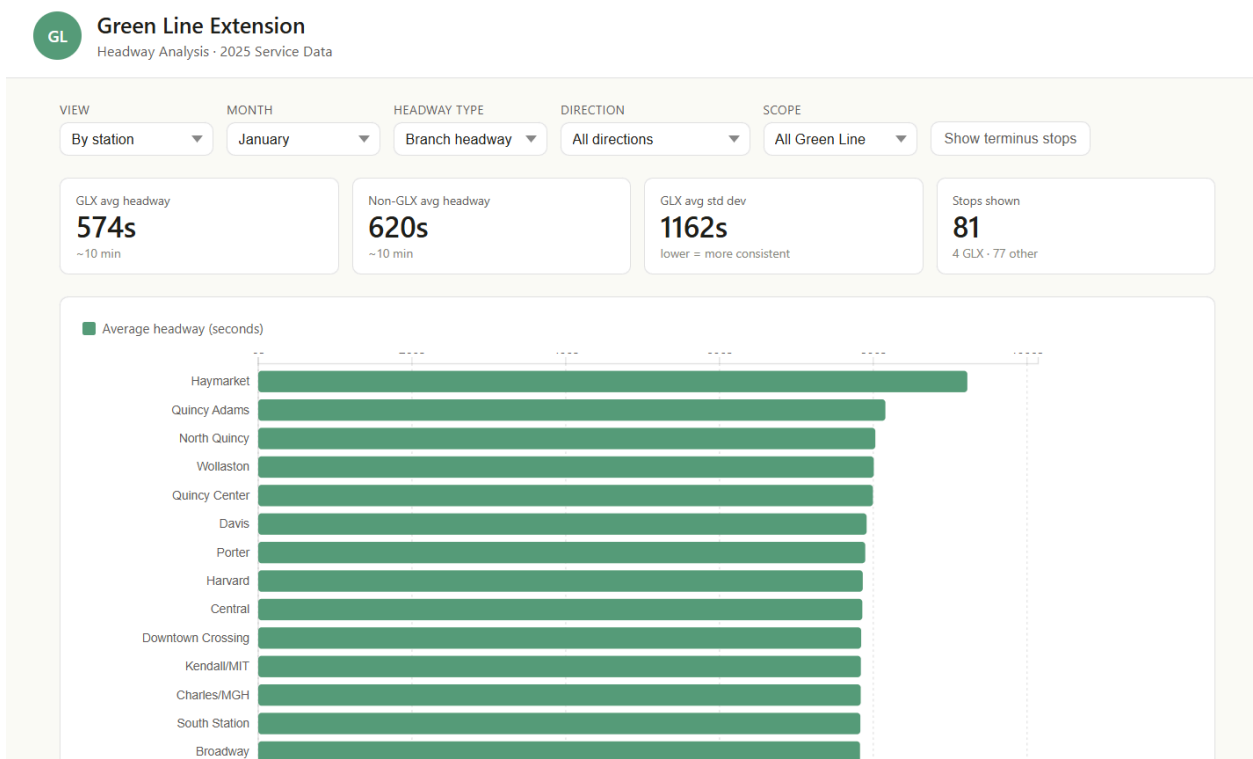


Figure 2: Average headway by station

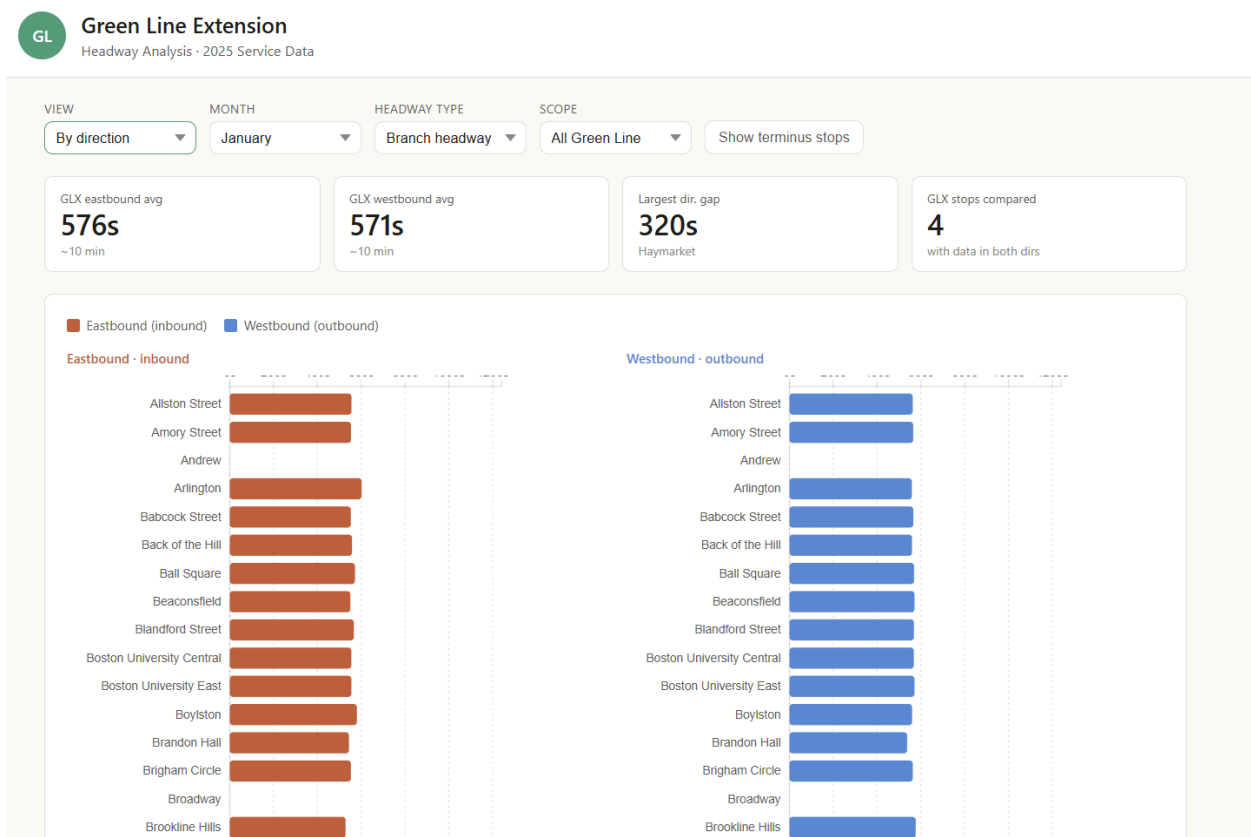
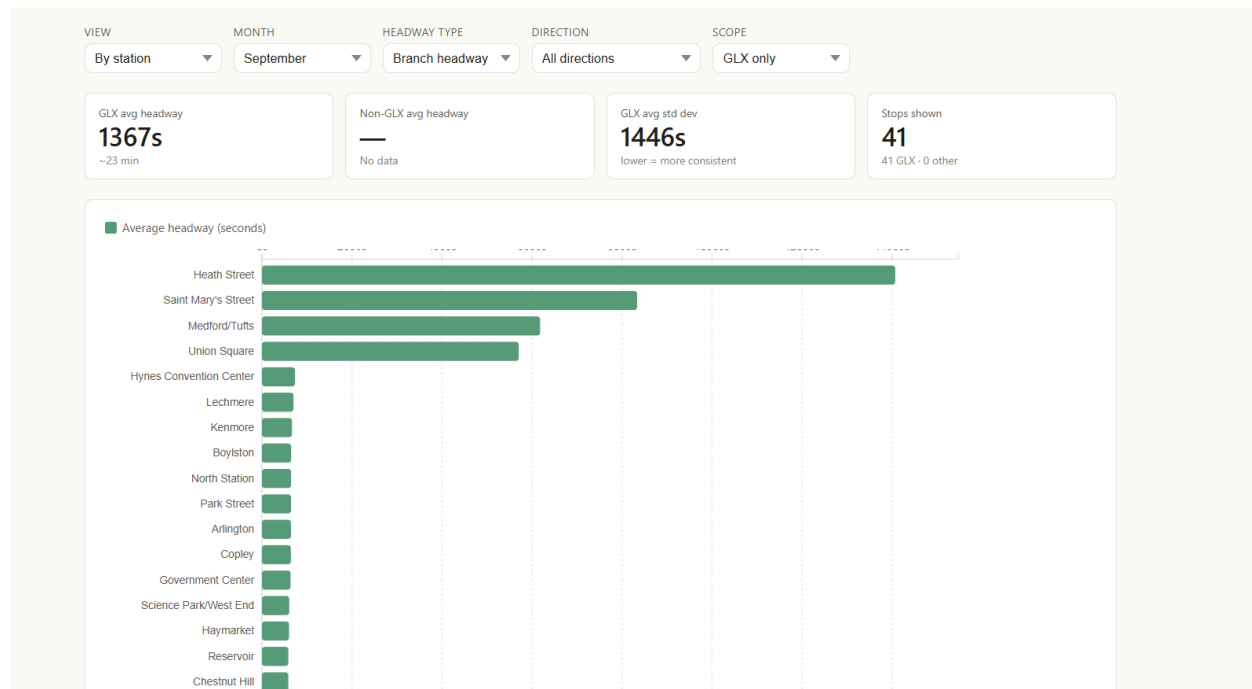


Figure 3: Average headway by direction per station

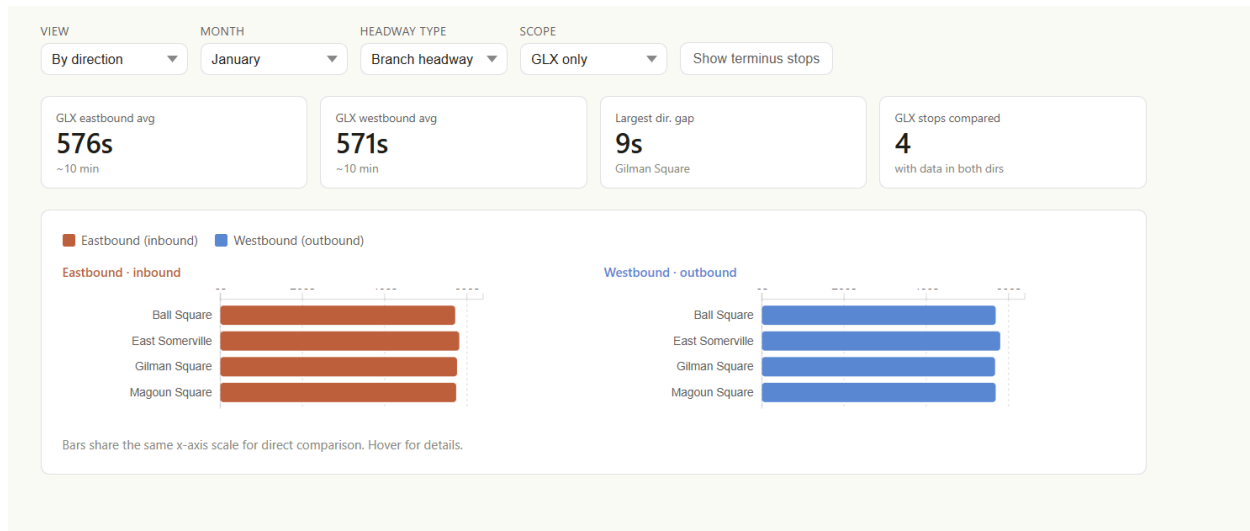
Question 2: Using this data set try to identify relationships between headways and the other variables available to you. Make an argument for why the relationships you find might exist. Please limit your answer to 8 images and 500 words.

The most pronounced relationship between headways and any other variable when first observing the data is that between it and stations, particularly between stations at the end of the line. In our visual analytics system, we have a feature that allows the user to filter out terminus stations as their headways were such outliers that it made it difficult to identify differences between other stations. This is seen in the figure below, with all of the extreme outliers being terminus stops.

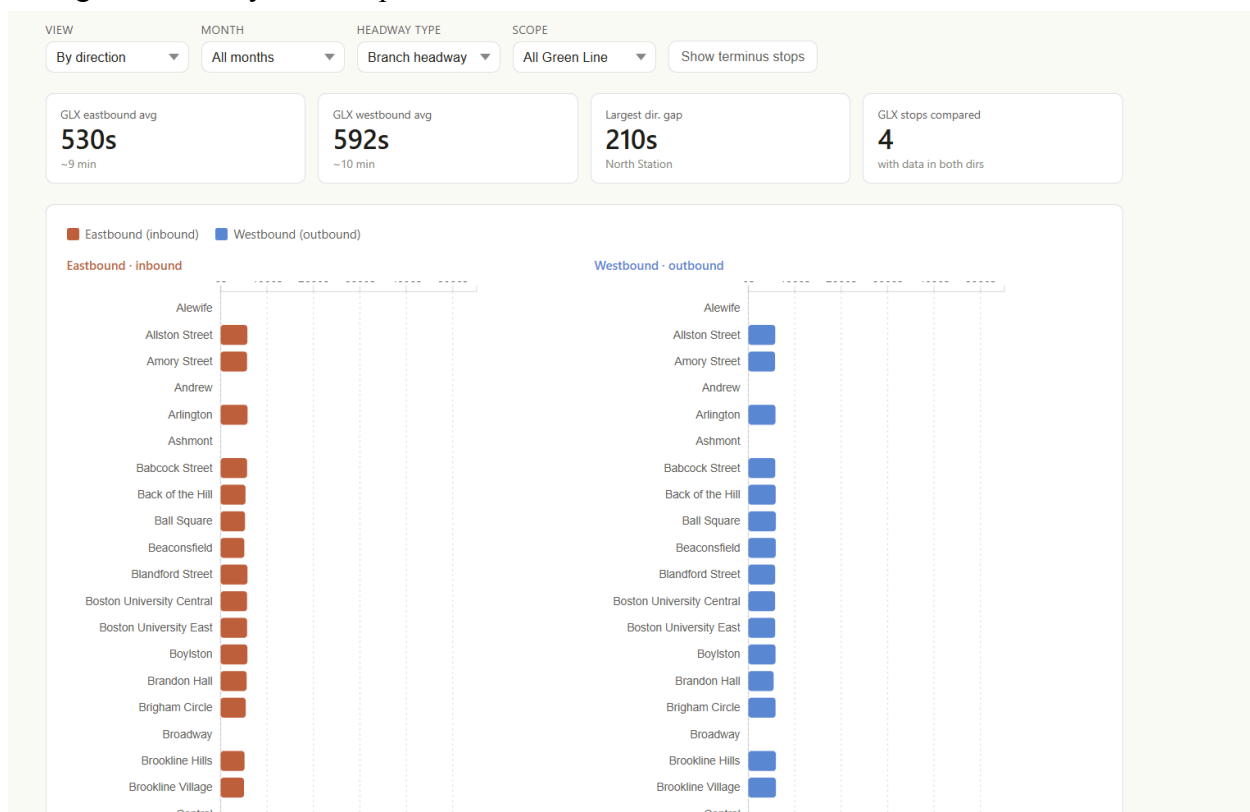


This relationship makes it very difficult to parse by station without excluding these stations, which we implemented a button to do. However, once we hide these terminus stops we can see that the green line extension stops are actually quite consistently timed. Additionally, there is no major difference between eastbound and westbound trains, the average being separated only by 5 seconds.

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However, when we look at the entirety of the list of stops in the dataset we find that this is actually a trait unique to the green line extension. Generally, the headway for eastbound trains is significantly lower, with the gap being as large as 210 seconds for North Station, which is one of the highest headway stations period.

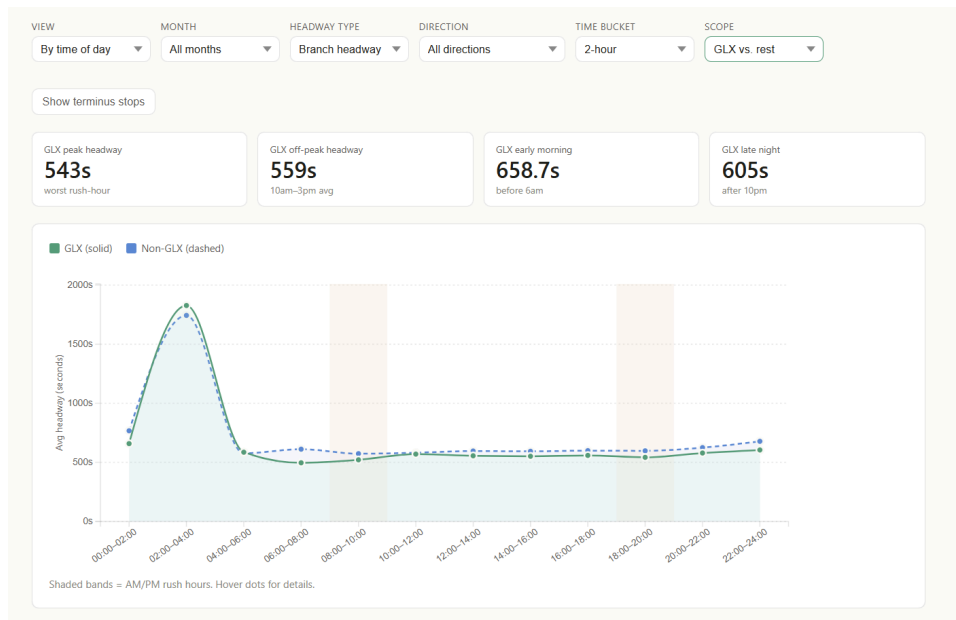


Thus, we feel that we can confidently say that the Green Line Extension stops are far more consistent by direction than other stops. Another interesting relationship we observed in the data

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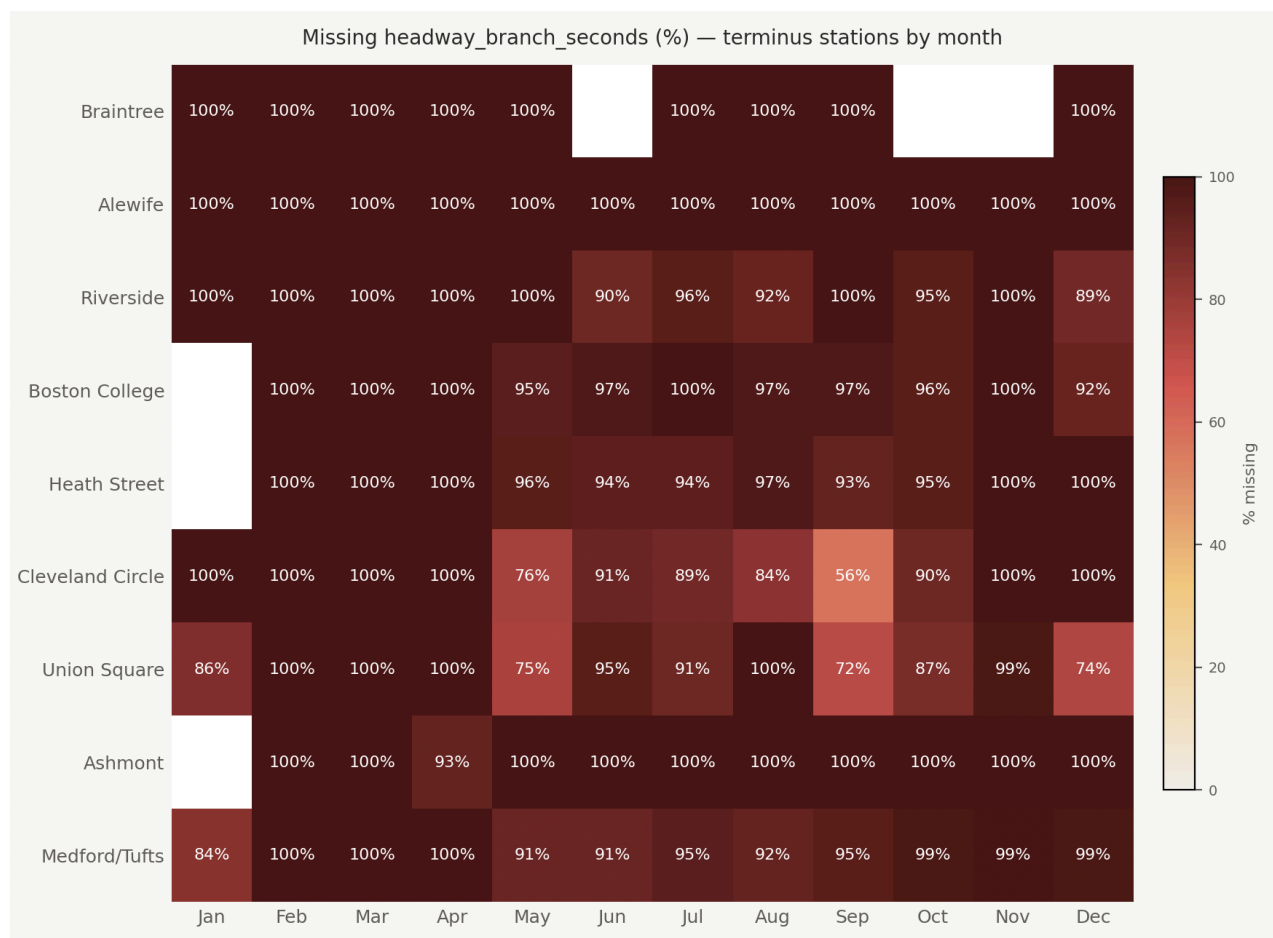
can be found by examining the headways by time of day for Green Line Extension and Non Green Line Extension stops. As seen in the figure below, the two groups follow roughly the same pattern. The Green Line Extension generally has slightly lower headways than other stops, this being most true just before the morning rush hour. We believe this may be due to the fact that GLX stops are distant from downtown, so their time of highest traffic and therefore service is a bit before rush hour in Boston commences. The only time that the Green Line Extension has higher headways than other stations is in the early morning, which is a time of greatly reduced to no service. Additionally, this figure shows us that generally headways are slightly improved for all stops at both the morning and afternoon rush hour, likely due to increased service, and are by far the worst at late night and early morning hours.



Overall, our visual analytics system allows the user to find relationships between headways and the time of day, direction of train, and by just the station with several options such as time buckets and disabling terminus stops to enhance this analysis.

Question 3: What are potential issues or shortcomings of this dataset? Consider how the data was collected. Are there any places in the data you think there might be errors? Provide evidence for these relationships.

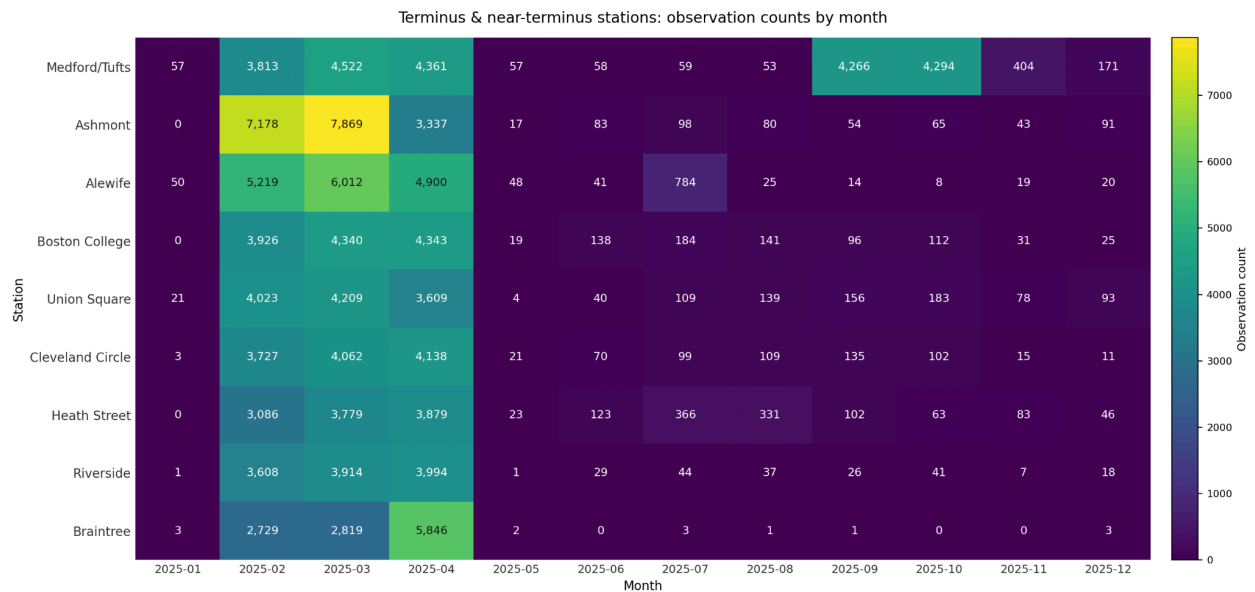
A key shortcoming of this dataset is the amount of missing data. About 31% of headway_branch_seconds values and 29% of branch_route_id values are missing. Headway_branch_seconds represents the time between consecutive vehicles leaving the same parent station on the same branch. The terminus is where branch headways should be the most complete and most informative because these values tell us how often trains are actually being dispatched onto the line. However, since the majority of the missing headway_branch_seconds occurs near terminus stations, we cannot see how often trains leave the start of each branch, which limits our ability to evaluate movement of trains along the line. If the data were complete, we would have a much clearer picture of service from the point of origin rather than only mid-line segments where headway statistics tends to be more reliable.

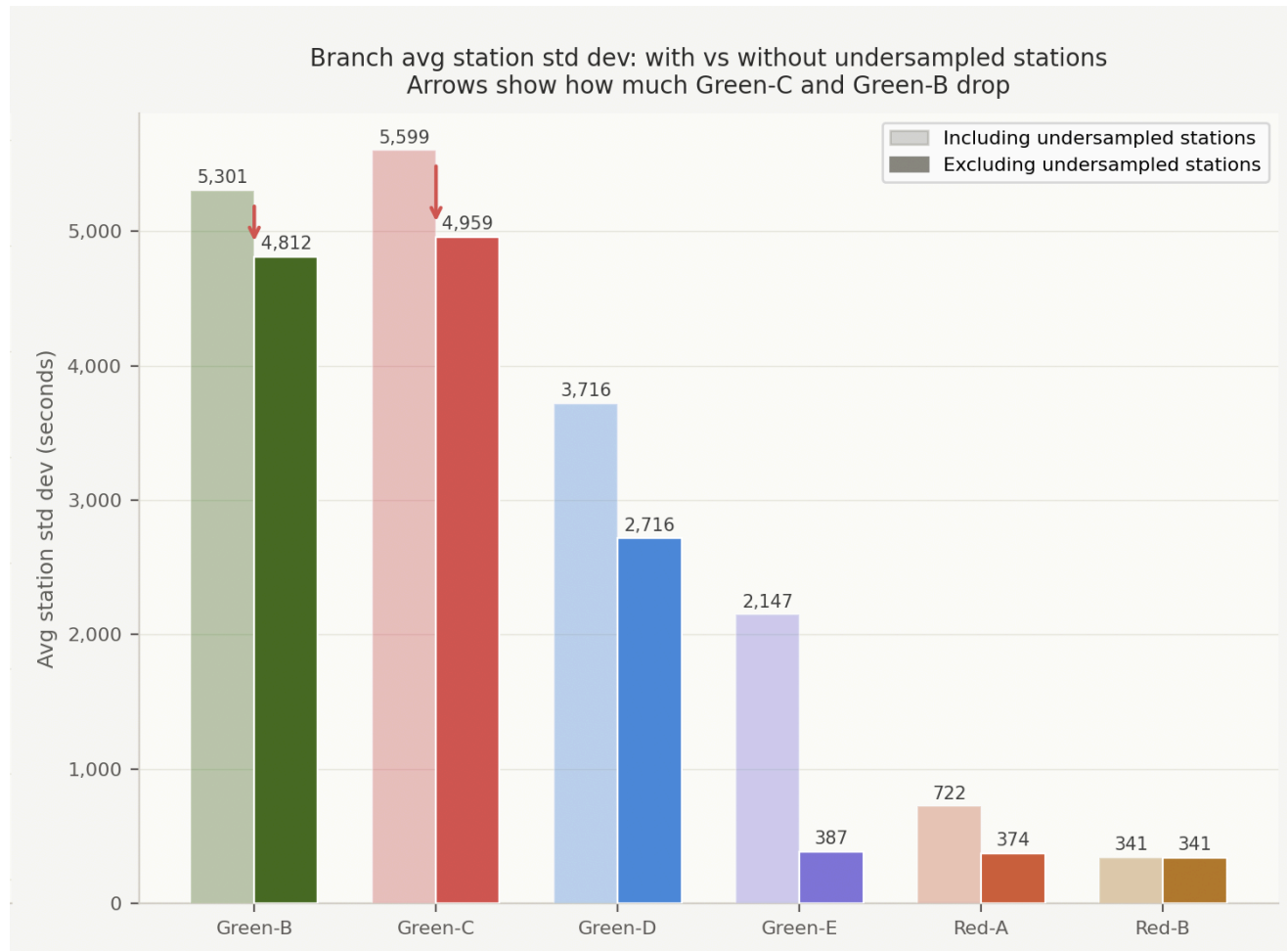


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Missing `branch_route_id` creates a similar issue, where it prevents us from assigning trips to specific branches on lines with multiple routes, such as the Green Line. Without this field, we cannot tell whether a vehicle belongs to the B, C, D, or E branch. These fields are important because they allow us to measure branch headways and identify which branch a vehicle belongs to. Without them, the dataset becomes biased toward observations where all data was successfully recorded.

Another limitation is uneven and very low sampling at certain stations. Branch terminus stations such as Boston College, Riverside, Union Square, and Medford Tufts sometimes have as few as 4 to 57 observations in a month. This is far lower than expected based on scheduled service. Since headways rely on consecutive observations, missing data points can make headways appear much larger than they really are. When these stations are included in branch level statistics, they can greatly increase the measured variability. This makes some branches, such as Green C and Green B, appear less reliable than they actually are.





This pattern can be seen in the station level variability plot. When stations with very low sampling are removed, the average standard deviation drops significantly. For example, the Green E branch decreases from 2147 seconds to just 38 seconds. This shows that much of the variability comes from missing data rather than actual service problems.

Finally, the dataset does not include important contextual information. It is missing variables such as scheduled headways, delay indicators, dwell times, passenger demand, and service disruption flags. Without this information, it is difficult to tell whether a large headway is caused by a real delay or simply missing data.